The Compound Monitor: Algorithm and Mathematical Machinery

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Abstract

A complete account of what the monitor computes, in the order it computes it. The mathematics is elementary: an AR(1) fit, a cumulative sum, two resampling procedures and a counting argument. What is not elementary, and what the document is really about, is *where the alarm threshold comes from.* Every constant is labelled as a convention, a free choice, a consequence of the method, or a value set by measurement. Section 10 gives the measured results and Section 11 the full constant table.

Contents

1 Objects and notation **2**
1.1 The loader boundary . . . 2

2 What changed: the before/after comparison **3**

3 Phase I: the baseline model **3**
3.1 Why a time-series model is needed at all . . . 3
3.2 Fitting . . . 4

4 The detector **5**
4.1 Choosing k . . . 5

5 The threshold **5**
5.1 Calibration A: the own-baseline bootstrap (the comparator) . . . 6
5.2 Calibration B: the reference library (the default) . . . 6
5.3 What the library can and cannot buy . . . 7

6 Is it working? The on/off trial **7**
6.1 The randomisation test . . . 7
6.2 Dropped cycles . . . 8
6.3 The verdict . . . 8

7 The three gates **9**
7.1 A precondition before any probability . . . 9
7.2 p_{complete}: will the trial finish? . . . 9
7.3 m_{det}: would a real effect be found? . . . 10

8 Planting **10**

9 The acceptance checks **11**
9.1 Check 11: the free diagnostic . . . 11

10 Measured results **11**

11 The constants **13**

1 Objects and notation

The subject. One person is a triple (id, x, t_0) where x is a real-valued series on a *complete* daily index: every calendar day between the first and last observation appears as a row, and a day with no reading is NaN rather than an absent row. The engine enforces this at construction. t_0 is the day the compound starts. Days before t_0 are the *before period*; days from t_0 on are the *monitoring period*.

Completeness of the index matters because the AR(1) model of Section 3 is a statement about *consecutive calendar days*. If missing days were dropped rather than marked, two readings a week apart would be treated as adjacent and the fitted autocorrelation would be wrong.

The metric. A metric is a tuple

$$(\texttt{name},\ \texttt{unit},\ \texttt{works_direction},\ \texttt{harm_direction},\ \texttt{onset_days}),$$

with the two directions in $\{\texttt{up}, \texttt{down}\}$. Write

$$s_{\mathrm{harm}} = \begin{cases} +1 & \texttt{harm_direction} = \texttt{up} \\ -1 & \text{otherwise} \end{cases} \qquad s_{\mathrm{works}} = \begin{cases} +1 & \texttt{works_direction} = \texttt{up} \\ -1 & \text{otherwise.} \end{cases}$$

The demo metric is resting heart rate in bpm, with $s_{\mathrm{works}} = -1$ (a recovery compound should lower it) and $s_{\mathrm{harm}} = +1$, and an onset of 2 days.

Nothing in the engine refers to heart rate. It sees a series and a metric. These four fields and the two signs are the only metric-specific quantities in the entire pipeline, which is why the same code runs on sleep efficiency without modification.

1.1 The loader boundary

The subject object above is the entire interface between the engine and a dataset. `engine.py` refers to no file format, no directory layout and no particular metric: it sees a series and a metric. Everything dataset-specific lives in a loader, so adding a dataset means writing a reader, not touching any of the mathematics that follows.

What a loader must produce. One value per person per day; a start date; and *a complete daily index*, with NaN for days that have no reading.

The third requirement is the one that bites, and it fails silently. A loader that simply omits missing days still runs, and the series still looks reasonable, but its index is no longer a run of consecutive calendar days. Equation (3.1) then pairs readings that are a week apart as though they were adjacent, $\hat{\phi}$ is wrong, and nothing anywhere raises an error.

For that reason the adapter layer exposes a single constructor, `to_subject()`, which every loader is expected to end in. It maps sentinel values to NaN, rejects duplicate dates rather than silently keeping one, reindexes to the complete daily range, and refuses a before period shorter than `MIN_BEFORE`. A loader that uses it cannot make the mistake above.

What does not transfer. Running is not the same as being calibrated. Several constants in Section 11 were set by measuring *this* cohort's characteristics rather than its file format: `MIN_USABLE_DAYS` from its missingness, and `REF_BASELINE_DAYS`, `HORIZON` and `BLOCK_DAYS` from records of about 150 days with ϕ between 0.58 and 0.92. The measured results of Section 10 belong to this cohort, this season and this device.

So a new dataset is loaded with a report of what the engine is about to assume about it — series length, coverage, longest gap, $\hat{\phi}$, n_{eff}, and the scorable-residual count over one horizon — and a reminder that checks 6, 7, 10 and 11 must be re-run to obtain its own numbers. The threshold itself additionally requires at least two subjects, since the default limit of Section 5.2 is built from other people; with one subject only the own-baseline comparator is available, and that is the construction measured at 14%.

A guard rather than a footnote. Because the failure mode here is a tool that displays a correct-looking rate belonging to different data, the measured results carry a fingerprint of the cohort they came from: subject ids, record spans, value counts and start dates. The display refuses to quote a false-alarm rate whose fingerprint does not match the data currently loaded, and says so instead. Portability without that guard would make the tool easier to misuse, not easier to use.

Three questions. The monitor answers, for one person:

1. **What changed?** Before versus after (Section 2).
2. **Is it harming them?** A sustained move away from their own normal, flagged automatically (Sections 3–5).
3. **Is it working for them?** An on/off trial (Section 6).

2 What changed: the before/after comparison

Let b be the non-NaN values before t_0 and a those from t_0 on (optionally truncated to a window). With $\bar{b}, \hat{\sigma}_b$ the sample mean and standard deviation (ddof $= 1$) of b:

$$\text{change} = \bar{a} - \bar{b}, \qquad \text{standardised change} = \frac{\bar{a} - \bar{b}}{\hat{\sigma}_b}.$$

Dividing by the *before* SD is what makes the number portable: "+2.1" reads as two of this person's usual swings, independent of units and of how variable they happen to be.

This quantity is displayed with a fixed caveat, because it is the one calculation here that cannot support a causal claim:

> A before/after difference alone can't show the compound caused it; season, training or illness could. The on/off trial is what attributes change to the compound.

Section 10 quantifies exactly how badly this caveat is needed: on data where nobody took anything, a third of monitoring windows show a shift in the mean.

3 Phase I: the baseline model

The framework is statistical process control. Phase I characterises "in control" from a reference period, here the before period; Phase II monitors new observations against it.

3.1 Why a time-series model is needed at all

Resting heart rate today predicts resting heart rate tomorrow. Every control-chart method assumes independent observations, so the standard remedy is to fit a time-series model and monitor its residuals. The relevant magnitude is the following.

Theorem 3.1 (Effective sample size)**.** *For a stationary AR(1) series with marginal variance σ^2 and autocorrelation ϕ,*

$$\delta(\bar{y}) = \frac{\sigma^2}{n^2}\left(n + 2\sum_{k=1}^{n-1}(n-k)\phi^k\right) \xrightarrow[n\to\infty]{} \frac{\sigma^2}{n}\cdot\frac{1+\phi}{1-\phi},$$

so the series carries the information of $n_{\text{eff}} = n(1-\phi)/(1+\phi)$ independent observations.

Proof. $\delta(\bar{y}) = n^{-2}\sum_{i,j}\operatorname{Cov}(y_i, y_j) = n^{-2}\sigma^2\sum_{i,j}\phi^{|i-j|}$. Counting the double sum by diagonals gives n terms with $|i-j| = 0$ and $2(n-k)$ terms for each $k = 1, \ldots, n-1$. Dividing by n and letting $n \to \infty$,

$$1 + 2\sum_{k=1}^{n-1}\left(1 - \tfrac{k}{n}\right)\phi^k \longrightarrow 1 + \frac{2\phi}{1-\phi} = \frac{1+\phi}{1-\phi},$$

by dominated convergence, since $\sum \phi^k$ converges. □

Measured on this dataset, ϕ ranges from 0.58 to 0.92 across participants. At $\phi = 0.85$, $n_{\text{eff}}/n = 0.082$: a 60-day baseline carries about five independent observations, and treating those 60 days as independent understates standard errors by $\sqrt{12.2} \approx 3.5$.

Remark 3.1 (The asymptotic formula fails for short blocks). The limit in Theorem 3.1 requires $n \gg (1-\phi)^{-1}$. At $\phi = 0.85$ that is $n \gg 6.7$, so for a block of $m = 5$ days the exact sum must be used. Exactly, $\delta(\bar{y}) = 0.785\,\sigma^2$, giving $\mathrm{SD}(\bar{y}) = 0.89\,\sigma$, whereas the asymptotic expression gives $1.56\,\sigma$ and overstates the noise by a factor of 1.76. This matters in Section 7, where blocks are five usable days long.

3.2 Fitting

From the before period, keeping NaN rows so that adjacency is preserved, let P be the set of present days.

Location and scale.

$$\hat{\mu} = \frac{1}{|P|}\sum_{t\in P} x_t, \qquad \hat{\sigma}_x^2 = \frac{1}{|P|-1}\sum_{t\in P}(x_t - \hat{\mu})^2.$$

If $|P| <$ `MIN_BEFORE` $= 14$ the engine refuses to fit.

Autocorrelation, with gaps. Let $A = \{t : t, t-1 \in P\}$ be the consecutive-present pairs. Then

$$\hat{\phi} = \mathrm{clip}_{[-0.95,\,0.95]}\left(\frac{\dfrac{1}{|A|}\displaystyle\sum_{t\in A}(x_t - \hat{\mu})(x_{t-1} - \hat{\mu})}{\dfrac{1}{|P|}\displaystyle\sum_{t\in P}(x_t - \hat{\mu})^2}\right). \tag{3.1}$$

This is not the textbook lag-1 autocorrelation and the difference is deliberate: the numerator averages only over pairs that genuinely are one day apart, while the denominator uses every present day, because every present day carries information about the variance. On a gapless series the two agree up to an $O(1/n)$ edge term. The clip keeps the model stationary when a short, noisy baseline produces $\left|\hat{\phi}\right|$ near or above one.

Residuals, then their scale, then standardisation. The order matters and is fixed:

$$e_t = (x_t - \hat{\mu}) - \hat{\phi}\,(x_{t-1} - \hat{\mu}), \qquad \hat{\sigma}_e^2 = \frac{1}{|E|-1}\sum_{t\in E} e_t^2, \qquad z_t = \frac{e_t}{\hat{\sigma}_e}, \tag{3.2}$$

where $E = A$ is the set of t for which e_t is defined. If either x_t or x_{t-1} is missing then e_t is NaN; the first day of the series has no predecessor and is NaN too.

Under the fitted model the z_t are approximately independent with mean 0 and variance 1, which is what the rest of the pipeline assumes. Whether that survives contact with real post-start data is measured, not asserted: see Section 9.1.

Carrying the fit forward. The monitoring period is standardised with the *same* $\hat{\mu}, \hat{\phi}, \hat{\sigma}_e$ fitted on the before period, applied along the whole series, so the first monitored day uses the last before-day as its predecessor and no day is wasted at the join.

Valid steps. For any window, write

$$n_{\text{valid}} = \#\{t \in \text{window} : z_t \neq \text{NaN}\}.$$

Because e_t needs two consecutive present days, if days are missing roughly at random with rate q then $n_{\text{valid}} \approx (1-q)^2$ times the calendar length. At $q = 0.34$ a 30-day window supplies about 13 valid steps, not 30. This number drives the calibration in Section 5 and is the single most frequent source of silent error in a design like this.

4 The detector

Definition 4.1 (One-sided CUSUM)**.** Given standardised residuals z_t, a reference value $k > 0$ and a sign $s = s_{\text{harm}}$,

$$S_0 = 0, \qquad S_t = \max(0,\ S_{t-1} + s\, z_t - k)\,, \qquad \text{alarm when } S_t > h.$$

If z_t is NaN then $S_t = S_{t-1}$ and no step is taken.

S_t is a random walk with a reflecting barrier at the origin. The barrier is what makes it a monitor rather than an accumulator: evidence that does not persist is forgotten.

Proposition 4.1 (The hypotheses are separated by the sign of a drift)**.** *If $\mathbb{E}[sz_t] < k$ the increments have negative mean, the walk is absorbed at the barrier and returns to 0 infinitely often. If $\mathbb{E}[sz_t] > k$ the increments have positive mean, and by the strong law $S_t/t \to \mathbb{E}[sz] - k > 0$, so S_t grows linearly and crosses any fixed h in expected time $O(h)$.*

So k is not a threshold but a *tilt*: it is chosen so that in control the walk drifts down and out of control it drifts up, and detection reduces to detecting the sign of a drift. This is exactly why the detector sees what a per-day limit and a self-report both miss. A long run of mildly elevated days climbs to the same place as one dramatic day, and nobody files a report for feeling slightly worse for three weeks.

4.1 Choosing k

The standard rule is to set k at half the shift one wants to detect, measured *in the units being monitored*. The units being monitored are residual standard deviations, while the target shift Δ is expressed in raw baseline SDs, so the two are related by the following attenuation.

Proposition 4.2 (Whitening attenuates a step shift)**.** *A sustained shift of $\delta\sigma_x$ in the mean of an AR(1) series appears in the standardised residuals as*

$$\delta\sqrt{\frac{1-\phi}{1+\phi}}.$$

Proof. Once the shift has persisted for more than one day, both x_t and x_{t-1} are displaced by $\delta\sigma_x$, so by (3.2) the residual retains $(1-\phi)\delta\sigma_x$. Dividing by $\sigma_e = \sigma_x\sqrt{1-\phi^2}$,

$$\frac{(1-\phi)\delta\sigma_x}{\sigma_x\sqrt{(1-\phi)(1+\phi)}} = \delta\sqrt{\frac{1-\phi}{1+\phi}}. \qquad \square$$

Hence

$$k = \frac{\Delta}{2}\sqrt{\frac{1-\hat\phi}{1+\hat\phi}}, \qquad \Delta = 2.0, \tag{4.1}$$

computed per person from that person's own $\hat\phi$. At $\hat\phi = 0.85$ the attenuation factor is 0.28, so $k \approx 0.28$ and a 2σ raw shift arrives as about 0.56 in residual units.

Remark 4.1 (The price, and a prediction that was wrong). Proposition 4.2 says the detector sees only 28% of the shift it would see on independent data. That is the accepted price of a valid false-alarm rate, and an invalid false-alarm rate is not a weaker guarantee but no guarantee at all. It is worth recording that the author predicted whitening would *cost power* relative to an unwhitened chart, and it does not: the null is attenuated by the same factor, so the signal-to-noise ratio is unchanged. Only the ratio matters.

5 The threshold

Let

$$M = \max_{1 \le t \le n_{\text{valid}}} S_t$$

be the running maximum over one monitoring course. We need

$$h = F_M^{-1}(1-\alpha), \qquad \alpha = \texttt{FP} = 0.05,$$

the $(1-\alpha)$ quantile of the null distribution of M. There is no closed form. **Where F_M comes from is the entire design question of this project**, and the false-alarm rate is the only quantity the project really measures.

Proposition 5.1 (The horizon must match)**.** *F_M depends on the number of* scored steps*, and M is nondecreasing in that number, since the maximum is taken over a larger set. A limit calibrated for n_1 steps and applied over $n_2 > n_1$ over-alarms; applied over $n_2 < n_1$ it under-alarms.*

This is why n_{valid}, not the calendar length, is passed to every calibration routine. A person with 34% missing days takes about 13 steps in a 30-day window, and a threshold built for 30 steps would make them look well-calibrated while silently costing sensitivity — on exactly the subjects whose data is worst.

5.1 Calibration A: the own-baseline bootstrap (the comparator)

1. Take the standardised before-period residuals z^{base}, NaNs dropped.
2. Draw n_{valid} of them with replacement; run Definition 4.1; record $\max_t S_t$.
3. Repeat $N = 1000$ times. Set h to the 95th percentile of those maxima.

Because Section 3 has already whitened the series, resampling single residuals with replacement is the ordinary residual bootstrap; no block structure is needed here.

This calibration needs nobody but the subject, which is its only advantage.

Proposition 5.2 (Why it fails)**.** *The procedure computes a quantile of $f_{H_0}^{\text{assumed}}$, the null built from the before period. The monitoring days follow $f_{H_0}^{\text{true}}$. If the mean shifts between the two periods these are different distributions, and the $(1-\alpha)$ quantile of one is not that of the other. Increasing N converges to the quantile of the wrong distribution.*

The error is therefore bias, not noise, and it has a direction: the assumed null contains none of the post-baseline drift, so it is too tight, h is too low, and false alarms exceed nominal. Measured: 14.0% against a nominal 5%.

5.2 Calibration B: the reference library (the default)

Replace the assumed null with a real one, built from other people who took nothing.

Building the library. For each subject j other than the target, slide through j's post-start series *in valid-step space*: for every index i whose residual z_i is defined and which lies at or after j's own start,

1. **Rebase.** Form the subject restricted to the `REF_BASELINE_DAYS` $= 60$ calendar days ending at $i-1$, with start i.
2. **Re-fit.** Fit $\hat{\mu}, \hat{\phi}, \hat{\sigma}_e$ on that 60-day window. If it holds fewer than 14 present days, skip this window.
3. **Score.** Compute residuals from day $i-1$ onward, standardise, drop NaNs and keep the first `HORIZON` $=$ 30 of them.

One window is produced per valid post-start day. On this dataset that is 876 windows from 13 subjects.

The threshold. Given the target's n_{valid} and k: take every library window belonging to another subject that supplies at least n_{valid} residuals, truncate each to its first n_{valid}, run the CUSUM *with the target's own* k, and set h to the 95th percentile of the resulting maxima. The statistic is therefore "what would *this* detector have done on *that* person's unexposed data," which is the right null for the question being asked.

Leaving the target out is not optional. Including their own data would fit the decision rule to the test data and the error estimate would mean nothing.

Why the baseline is re-fitted per window. A single fixed baseline would score a window 90 days later against a 90-day-old reference, which no real course does. Worse, most subjects in this dataset share a start date, so the earliest windows all cover the same calendar stretch; sliding one step at a time then replicates that one shared drift episode into hundreds of overlapping windows until the 95th percentile lands inside it. Measured: the demo subject's h rises from 8.5 to 16.9 and detection falls to 21%. Re-fitting per window removes it.

5.3 What the library can and cannot buy

Proposition 5.3 (A quantile from a small sample)**.** *The sample $(1-\alpha)$ quantile from m independent draws from a density f has asymptotic variance $\alpha(1-\alpha)/(mf(q)^2)$, which is large for small m and larger still in a tail where $f(q)$ is small. At $\alpha = 0.05$ and $m = 13$,* $\Pr(\textit{all 13 draws lie below } q) = 0.95^{13} = 0.51$*: about half the time the entire sample sits below the quantity being estimated.*

Overlapping windows do not increase m. Windows from one person share almost all their data, so the effective sample size is the *subject* count, not the window count. Both are reported everywhere.

Remark 5.1 (Measured). Recomputing h with each library subject left out in turn moves it by 10% to 46% of its value (median 37%); for the demo subject the range is 5.75 to 8.68. One subject dominates: 22 of the 30 largest library maxima are theirs, and dropping them lowers h for 10 of 13 subjects.

Yet the false-alarm *rate* barely moves: 7.0% for twelve of the thirteen removals, 11.6% for the one. The threshold swings across a gap in the null's order statistics where few observed maxima lie, so a large change in h produces a small change in the rate. Both numbers are reported, because the first is the measured reason a deployment needs a reference cohort of 50 to 100 people rather than 13, and enlarging the library cannot substitute: going from 60 to 500 windows moved h by 0.4.

6 Is it working? The on/off trial

The safety arm needs no assumption about the compound. The efficacy arm needs to attribute a change to it, which requires a design rather than an analysis.

The design. n cycles. Each cycle is two adjacent blocks of `BLOCK_DAYS` $= 7$ days, one on and one off, with the order inside the cycle decided by a coin flip. The first `onset_days` of every block are excluded from analysis as washout, since a compound with a half-life is still present after a block ends. With a 2-day onset, each block contributes five usable days.

The statistic. For each cycle with both blocks usable (at least `MIN_USABLE_DAYS` $= 2$ post-washout days each), let

$$d_c = \overline{(\text{on block})} - \overline{(\text{off block})},$$

the difference of post-washout means. Block means rather than individual days, because days are autocorrelated and treating them as independent inflates false positives; averaging within blocks is the standard n-of-1 remedy.

6.1 The randomisation test

Definition 6.1 (Randomisation p-value)**.** Let G be the set of label assignments the randomisation could have produced and $T(y, g)$ the statistic under assignment g. Then

$$p = \frac{\#\{g \in G : |T(y,g)| \geq |T(y, g_{\text{obs}})|\}}{|G|}.$$

Validity comes from the coin flips that were actually made. Under the null that the compound does nothing, the observed data would have been identical whichever assignment came up, so all $|G|$ values of T were equally likely and the rank of the observed one is uniform. **No normality, no independence and**

no stationarity is assumed, so none of them can break it. This is why the efficacy arm is untouched by the $\phi \approx 0.85$ problem that dominates the safety arm.

Here the order is randomised *within each cycle*, so G is the set of sign patterns $\{\pm 1\}^n$ and $|G| = 2^n$.

Theorem 6.1 (Minimum attainable p)**.** *The map $g \mapsto g^c$ that flips every sign is an involution on G with $T(y, g^c) = -T(y, g)$. Hence $|T|$ is constant on complementary pairs, the observed assignment always ties with its own complement, the numerator is at least 2, and*

$$p_{\min} = \frac{2}{2^n}.$$

Corollary 6.2 (`MIN_CYCLES` $= 6$)**.** $p_{\min} < \alpha = 0.05$ *requires* $2^n > 40$*, hence* $n \geq 6$*. At* $n = 3$*,* $p_{\min} = 0.25$*; at* $n = 5$*,* 0.0625*; at* $n = 6$*,* 0.03125*.*

This is a statement about the cardinality of a finite set, not about power. No effect size, no amount of data and no better instrument changes it. Below six cycles the app refuses the schedule and says why, *before* it is run, because a tool that accepts a 3-cycle schedule, runs it for six weeks and reports $p = 0.25$ has produced a confident-looking null determined by the calendar rather than the compound.

Corollary 6.3 (The test is conservative by construction)**.** *The attainable p-values lie on the grid* $\{2/2^n, 4/2^n, \dots\}$*. At* $n = 6$*,* $p < 0.05$ *means* $p = 2/64 = 0.031$ *exactly, so the true false-positive rate is about* 3.1% *rather than* 5%*. Measured:* 3.6% *over* 140 *null runs.*

6.2 Dropped cycles

If some cycles are unusable the test runs on the survivors, and the implementation handles this by zero-padding rather than by re-indexing.

Lemma 6.4 (Zero-padding preserves the p-value)**.** *Let k of n cycles be usable and set $d_c = 0$ for the rest. Then the proportion of sign patterns in $\{\pm 1\}^n$ at least as extreme as the observed one equals the proportion over the 2^k patterns on the usable cycles alone.*

Proof. The statistic $\sum_c \varepsilon_c d_c$ does not depend on ε_c when $d_c = 0$. So each of the 2^k distinct values arising from the usable cycles is attained by exactly 2^{n-k} patterns of the full set. Both numerator and denominator are multiplied by 2^{n-k}, and the ratio is unchanged. □

The reported floor uses k, not n: a trial that was scheduled for eight cycles and kept seven is judged at $p_{\min} = 2/2^7$.

6.3 The verdict

With k usable cycles, $\bar{d}$ their mean and p the randomisation p-value:

$$\begin{cases} \textbf{Not enough usable cycles} & k < \texttt{MIN_CYCLES} \\ \textbf{Consistent effect: working} & p < \alpha,\ \operatorname{sign} \bar{d} = s_{\text{works}} \\ \textbf{Consistent effect: opposite to intended} & p < \alpha,\ \operatorname{sign} \bar{d} = -s_{\text{works}} \\ \textbf{No consistent effect detected} & \text{otherwise.} \end{cases}$$

The last is always shown with the sentence *"This is not the same as no effect. This schedule reliably detects effects of about m SDs or larger,"* with m from Section 7. An "opposite" verdict is displayed alongside the harm status, since a compound moving the metric the wrong way is a safety observation as much as an efficacy one.

7 The three gates

Before a schedule is run, three numbers are computed and shown. They answer three different questions, and conflating them is how a trial ends up producing a meaningless null.

quantity	question	section
p_{floor}	can a result exist at all?	6.1
p_{complete}	will the trial finish?	7.2
m_{det}	would a real effect be found?	7.3

7.1 A precondition before any probability

If the schedule runs past the last day with data, there is nothing to compute. A schedule of n cycles beginning at t_0 ends at $t_0 + 2n \cdot \texttt{BLOCK_DAYS} - 1$, and the engine raises an error naming how many cycles would fit instead. This is a length check of the same kind the loader already applies to the before period, and it is the right answer for a device that stopped recording: not "81% chance of completing" but "this schedule does not fit in the available record."

On this dataset, six of thirteen subjects are rejected here and seven fit.

7.2 p_{complete}: will the trial finish?

A block is dropped if it has fewer than `MIN_USABLE_DAYS` post-washout days with data; a cycle needs both of its blocks; the verdict needs `MIN_CYCLES` cycles. So

$$p_{\text{complete}} = \Pr\big(\text{at least } \texttt{MIN_CYCLES} \text{ of } n \text{ cycles survive}\big).$$

Not the probability that all survive. The verdict needs six usable cycles, not every scheduled one: schedule ten, lose three, and the test runs on seven. The all-must-survive version *decreases* as cycles are added, which would make "consider more cycles" actively harmful advice; the correct version increases. The two coincide exactly at $n = \texttt{MIN_CYCLES}$, which is why the error is invisible there.

The binomial version, and why it is not enough. If missing days were independent with coverage c, then with u post-washout days per block,

$$p_{\text{block}} = \Pr(\text{Bin}(u, c) \geq \texttt{MIN_USABLE_DAYS}), \qquad p_{\text{complete}} = \Pr\big(\text{Bin}(n, p_{\text{block}}^2) \geq \texttt{MIN_CYCLES}\big)\,.$$

Real missing days come in runs, and a run is far more destructive than the same number of scattered days: losing two days of five leaves the block usable, while losing five consecutive days destroys the block, the cycle and (without slack) the trial. The binomial cannot see this. Measured, the binomial predicted 0.06 to 0.90 for the two gap-prone subjects, and both reached a verdict in 0 of 20 real schedules.

The moving-block bootstrap. Instead of modelling the gap pattern, resample it. Take the subject's own before-period missingness indicator $m \in \{0,1\}^N$. For block length L, draw $\lceil T/L \rceil$ starting positions uniformly from $\{0, \ldots, N - L\}$, concatenate the blocks $m[s : s + L]$ and truncate to the trial length T. Lay the result over the planned schedule and count whether at least `MIN_CYCLES` cycles survive. Repeat 10,000 times.

Runs appear automatically because runs are what is being resampled, and the pattern is the subject's own rather than an assumed one.

The block length is not a free choice. L changes the answer: for one subject p_{complete} is 0.90/0.79/0.64/0.77 at $L = 3, 5, 7, 14$, and the warning fires at 0.8. A gate whose output depends on an arbitrary constant is not a gate, so the engine computes all four and **reports the minimum**. That is the conservative end of a sensitivity analysis rather than a tuned value: a warning should fire if any defensible assumption says it should. All four are shown in the app.

The non-monotonicity in L is itself informative. It means that subject's gaps have structure at a particular scale rather than being random runs, and 7-day blocks happen to reproduce it.

If the before period is shorter than `MIN_RESAMPLE_DAYS` $= 28$ there is too little pattern to resample and the binomial is used, flagged as optimistic.

One standing caveat. p_{complete} is conditional on the device continuing to record as it has been. Nothing can predict a watch that stops.

7.3 m_{det}: would a real effect be found?

p_{floor} says a result is *possible*; this says whether it is *likely*. Simulate the whole pipeline:

1. Generate N trials of length $T = 2nB$ from the fitted AR(1) with unit marginal variance: $x_1 \sim N(0,1)$ and $x_t = \hat{\phi} x_{t-1} + \varepsilon_t$, $\varepsilon_t \sim N(0, 1-\hat{\phi}^2)$. Unit marginal SD means an effect of δ is δ baseline SDs by construction.
2. Apply a missingness draw from Section 7.2, at the block length that produced the minimum.
3. Randomise the on/off order within each cycle.
4. Add $s_{\text{works}}\,\delta$ on the on-blocks, ramped linearly over `onset_days` at the start of each on-block — the same shape `plant_effect` uses, so "would be found" and "was found" cannot contradict each other.
5. Compute block means, cycle differences, the usable-cycle mask and the exact randomisation p. A trial counts as detected when it has at least `MIN_CYCLES` usable cycles, $p < \alpha$, *and* the mean difference has the intended sign.
6. Sweep δ over $0.1, 0.2, \ldots$ and report the first reaching 80%.

The plateau rule. Power can never exceed p_{complete}, since a trial that does not finish cannot detect anything. When that ceiling sits within two standard errors of the target,

$$p_{\text{complete}} < 0.80 + 2\sqrt{\frac{0.8 \times 0.2}{N}} \approx 0.825 \quad (N = 1000),$$

the power curve is flat near the target and any crossing is a draw from a flat region. The engine then reports **not reachable** rather than a number, because quoting 2.4 when 2.8 and 3.2 are equally consistent implies a precision that does not exist.

8 Planting

PMData participants took nothing, so there is no real effect in the data. Two functions inject one, and they are the only way an effect enters.

Harm. From $t_0 + 5$ days, add $s_{\text{harm}}\,\delta\,\hat{\sigma}_x$ to every value, ramped linearly over three days and sustained thereafter. NaN days stay NaN.

Effect. Add $s_{\text{works}}\,\delta\,\hat{\sigma}_x$ during on-blocks only, ramped over `onset_days` at the start of each on-block, matching the washout exactly so that the ramp falls entirely inside the excluded days.

Both return a new subject with `simulated` = True, a flag that survives every transformation and drives the on-screen label. The two controls are mutually exclusive in the app, since the works and harm directions are opposite and planting both would partially cancel during on-blocks.

Remark 8.1 (Planting is a measurement, not a fabrication). Injecting a known signal to characterise an instrument is what calibration is: a smoke detector is tested with a known quantity of smoke. The property that makes it honest is that the answer is known before you look, so there is no room to interpret the outcome favourably. Note also the separation of concerns: the *false-alarm* rate is measured on entirely unmodified real data, where every alarm is false by construction, and planting enters only the *detection* measurement. Finally, the result is unflattering — 53% — which is what a measurement looks like and what a fabrication does not.

9 The acceptance checks

Eleven checks, run as one script. Some have pass conditions; three are measurements reported without a threshold, because a quantity you are measuring cannot also be a target you are hitting.

#	what it tests	kind
1	loader: complete index, no zeros, NaN accounting	pass/fail
2	before/after on a known +5 shift at SD 2	pass/fail
3	$\hat{\phi}$ recovery and residual whitening on synthetic AR(1)	pass/fail
4	CUSUM at rest: all-zero z gives all-zero S	pass/fail
5	bootstrap self-consistency: fresh windows alarm at 5%	pass/fail
6	**false alarms on real unexposed data, both limits**	measured
7	detection with a planted 2σ harm, both limits	pass/fail
8	the three gates, the length precondition, block sensitivity	pass/fail
9	efficacy wiring on a planted effect	pass/fail
10	**efficacy false positives on real data**	measured
11	**baseline-model diagnostic on real monitoring days**	measured

Check 5 deserves a note: it resamples from the same residuals the limit was built from, so it verifies the *code* and says nothing about real-world validity. It is checks 6 and 11 that do that.

9.1 Check 11: the free diagnostic

Everything downstream assumes the fitted $\hat{\mu}, \hat{\phi}, \hat{\sigma}_e$ still describe the person during monitoring. Check 11 tests that directly, with no detector, no simulation and no planting. On each real monitoring window, compute z_t and report the distribution across windows of

$$\bar{z} \quad \text{(should be 0)}, \qquad \mathrm{SD}(z) \quad \text{(should be 1)},$$

together with the share of windows where $|\bar{z}| > 2/\sqrt{n_{\text{valid}}}$, roughly two standard errors.

This is the free diagnostic: no labels, no held-out data, no detector output. It separates the two ways the model can fail, which nothing else in the build does. $\bar{z} \neq 0$ is a *location* error, the mean moved; $\mathrm{SD}(z) \neq 1$ is a *scale* error, $\hat{\phi}$ or $\hat{\sigma}_e$ was mis-estimated. The 14%-versus-7% comparison tells you the null is wrong but not which.

10 Measured results

PMData (Thambawita et al., Simula; CC BY 4.0), daily resting heart rate from 16 participants of whom 13 are usable, recorded November to March. Nobody took anything, so every alarm below is false by construction. Windows start at days $60, 74, 88, 102$, each with its own 60-day baseline; 43 windows from 13 subjects.

Safety arm.

	reference limit (default)	own baseline (comparator)
false alarms	7.0% (3/43)	14.0% (6/43)
Wilson 95% CI	2.4–18.6%	6.6–27.3%
subjects with an alarm	2/13	5/13
2σ harm caught within 30 d	53% (23/43)	79% (34/43)
days to alarm, median (IQR)	13 (10–22)	11 (8–17)

Intervals treat windows as independent; they overlap within a person, so the true intervals are wider.

The own-baseline limit catches more *because its h is lower*, which is the same fact as its 14% false-alarm rate rather than a separate benefit. These are one detector at two points on the same curve, not two methods.

Detection depends on the person's own drift. Splitting the 43 windows by their check-11 mean z:

window's own drift	planted harm caught	median day
toward harm ($\bar{z} > 0$)	14/15 (93%)	11
away from harm	9/28 (32%)	21

So 53% is not a per-person coin flip; it is a blend of near-certain and unlikely, and which applies cannot be known in advance. The split uses the same days the detector scores, so it describes outcomes rather than predicting them. It also cannot be turned into a runtime confidence qualifier: a window drifting toward harm and a window containing harm are the same observation, so $\bar{z}$ carries no information the detector does not already use.

Efficacy arm. False positives 3.6% (5/140 runs, Wilson 1.5–8.1%), against 3.1% expected from Corollary 6.3. Minimum detectable effect 1.7 to 1.9 baseline SDs for subjects whose schedule fits, about ±0.1 of Monte Carlo noise. Seven of thirteen subjects can run a six-cycle trial; none fits eight.

Check 11.

	measured over 43 windows	if the model held
SD(z)	median 0.97 (IQR 0.79–1.10), RMS 1.02	1
$\bar{z}$	median −0.09 (IQR −0.43 to +0.10), range −0.85 to +0.69	0
$\lvert\bar{z}\rvert > 2/\sqrt{n_{\text{valid}}}$	15/43 (35%), Wilson 22–50%	about 5%
lag-1 autocorr. of z	median +0.12 (IQR −0.08 to +0.27)	0

Read this as three statements.

The scale holds. $\hat{\phi}$ and $\hat{\sigma}_e$, fitted on 60 days before exposure, carry over to the monitoring period. This is the half of the model that had to work: a scale error would mean z is not comparable between people, so borrowing other people's CUSUM maxima would be borrowing numbers in different units and Section 5.2 would be invalid.

The location fails, with no common direction. The median $\bar{z}$ is essentially zero while the spread runs from −0.85 to +0.69. This is the measured answer to "why not just correct for the drift?": there is no common drift to subtract, so a population template would change nothing. Only a reference-group threshold helps, because it absorbs spread where a template removes a mean.

A small residual autocorrelation remains. The $2/\sqrt{n}$ bound assumes independent residuals, and z retains a lag-1 of +0.12, consistent with $\hat{\phi}$ being biased low by roughly $(1 + 3\phi)/n \approx 0.06$ from a 60-day fit. Widening the bound to allow for it still leaves 13 of 43 windows. A bias-corrected $\hat{\phi}$ is the measured next step; it is not implemented, and the screens state the measured 7% rather than the nominal 5%, so nothing is claimed that has not been measured.

11 The constants

name	value	status
`FP`	0.05	convention; false-alarm rate per course
`ALPHA`	0.05	convention; efficacy significance level
`N_SIMS`	1000	enough for a stable 95th percentile
`HORIZON`	30 d	choice; length of one course
`DELTA`	2.0	choice; shift the detector is tuned for
`MIN_BEFORE`	14 d	choice; refuse to fit below this
`BLOCK_DAYS`	7	choice; valid only for compounds acting within `onset_days`
`REF_BASELINE_DAYS`	60 d	choice; re-fitted per library window
`MISS_BLOCK_LENGTHS`	$3, 5, 7, 14$	choice; p_{complete} is the *minimum* over them
`MIN_RESAMPLE_DAYS`	28	choice; below this, fall back to the binomial
start day	60	choice (demo only; there is no real exposure)
plant sizes	2.0, 1.0	choice; always labelled
k	eq. (4.1)	**follows** from Δ, $\hat{\phi}$ and Prop. 4.2
`MIN_CYCLES`	6	**follows** from $p_{\min} = 2/2^n$ and `ALPHA`
`DEFAULT_CYCLES`	8	**follows**: two cycles of slack, so one bad week does not end a trial
plateau cutoff	0.825	**follows**: two standard errors above 0.80 at $N = 1000$
`MIN_USABLE_DAYS`	2	**measured**: at 3, completion is 0.49/0.24/0.05 at 20/26/34% missing, and dropping to 2 costs nothing on complete data
`N_SIMS_COMPLETE`	10,000	**measured**: ± 0.01 noise, so the 0.8 warning is not seed-dependent

12 What this does and does not establish

Established. Calibrating a per-person monitor from the subject's own pre-exposure baseline alarms on 14% of unexposed monitoring courses against a nominal 5%, and the cause is a location error with no common direction (Section 9.1). A leave-one-subject-out reference threshold brings this to 7% while still catching 53% of a sustained 2σ shift, typically by day 13. Both numbers are measured on real data from people who took nothing.

Not established, and stated as such.

- Transfer. The 7% and 53% belong to this cohort, this season and this device. Check 11 shows the location error is substantial and person-specific, so a new deployment must measure its own.
- Seasonality. Every participant was recorded November to March, so cohort-wide seasonal drift sits inside the empirical null instead of being caught by it. The 7% is optimistic for a cohort spread across different months, and this cannot be fixed with this dataset.
- Cohort size. With 13 subjects the threshold is effectively set by one person's drift episode (Section 5.3). A deployment needs 50 to 100.
- Trial length. The efficacy arm needs six cycles at minimum and eight by default, roughly 112 days plus a baseline. No subject here fits eight.
- Multiplicity. One primary metric per compound, fixed in advance. Monitoring four at a nominal 5% each gives $1 - 0.95^4 = 18.5\%$ per course.
- Portability. The engine runs on any dataset that can be turned into subjects (Section 1.1), and this has been exercised once already on a second metric with different units, a different direction convention and a different missingness pattern. But the constants and the measured numbers are this cohort's, and the display refuses to quote a rate it has not measured on the data in front of it.

Hypotheses tested and rejected. That AR(1) whitening costs power (it does not; the null attenuates identically). That a larger reference library lowers h (twelve times more windows moved it by 0.4). That drift could be removed with a population template on this dataset (there is no common drift to remove). Each was predicted, measured and abandoned.

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